Results & Discussions

1. Plot the engineering stress vs. strain curve for the annealed, normalized, water quenched, oil quenched, and two quenched-and-tempered specimens of 4340 steel on the same graph. Label each curve.

2. Calculate and tabulate the elastic modulus, yield strength, ultimate strength, toughness, and % elongation for each specimen. Make sure to use fractional strain to determine E. Also include the corrected Brinell hardness (BHN) for each specimen. Note that you can convert Rockwell to Brinell hardness using the chart in the lab manual or by the following approximate equations: *BHN = 33.22 exp(0.0192 HRB)* or *BHN = 131.7 exp(0.0252 HRC)*

3. Discuss how heat treatment (specifically cooling rate) affects mechanical properties. Which heat treatment produced specimens that were strongest? Hardest? Stiffest? Most ductile? Toughest?

4. Make a plot of σ_{uts} vs. BHN using data from all of these samples, and fit a straight line to the data. Include a "theoretical" line. Compare hardness values with ultimate strength for each specimen. Is this relationship what you expect?

5. State which heat treatment(s) may give rise to ultimate strength values that are affected by the populations of surface flaws. (*It may help to review your lab and notes from fracture*)

6. Using the oil quenched and both quenched-and-tempered specimens, plot the ultimate strength, yield strength, elastic modulus, and % elongation versus tempering temperature.

7. In your report, include micrographs of any three of the heat treatments that you were shown in the lab. Label the phases. Compare and contrast the resulting microstructures.

8. Using the heat treatment micrographs from lab, describe the microstructure that is present in each case, stating what phases are present and in (approximately) what amounts. When possible, compare the relative phase amounts to what you expect from the equilibrium iron-carbon phase diagram.

9. How does each microstructure arise from its heat treatment? Explain how material properties relate to these heat treatment procedures as well as the microstructures (i.e. how does processing effect microstructure effect properties?)